



PROFILE SUMMARY

Researcher in formal reasoning with a background in mathematics & computer science. My current research focuses on automated reasoning, SMT solving, and formal verification, particularly satisfiability certification for quantified formulas. More broadly, I am interested in the connections between mathematical logic, theorem proving, and automated methods for solving computational problems.

CONTACT DETAILS

@ nugraha@cs.cas.cz
+420 735 026 408

RESEARCH INTEREST

- Formal Verification
- Automated Reasoning
- Theorem Proving
- Logical Foundations of Computation
- Formalised Mathematics
- Foundations of Mathematics
- Philosophical logic

TECHNICAL SKILLS

- Language: Bahasa Indonesia, English
- Programming: Python (developing), Java, C
- Scientific Computing: Maple, Matlab
- Typesetting: LaTeX

EXPERIENCE

VĚDECKÝ ASISTENT (SENIOR RESEARCH ASSISTANT)
*Institute of Computer Science,
Czech Academy of Sciences.* **08.09.2025–present**

◇ Conduct research in automated reasoning, SMT solving, and AI methodologies within the Statistical Modelling and Artificial Intelligence Department. Concurrently advancing specialised knowledge through a secondary PhD program.

LECTURER (ASSISTANT PROFESSORIAL FELLOW 5)
*Institute of Mathematics
The University of Philippines–Diliman* **Aug 2024–Jul 2025**

◇ Designed and delivered comprehensive undergraduate lectures, developed course materials, and administered academic examinations.

CAREER INTERRUPTION **2019–2024**
◇ Academic career break devoted to religious vocation & monastic formation.

TUTOR
*School of Mathematics & Statistics
University of Canterbury, NZ* **2015–2018**
◇ Facilitated weekly tutorial sessions for groups of 20+ undergraduate students. Evaluated examinations and assignments, and served as an official proctor.

LECTURER AND RESEARCHER
*Faculty of Computer Science
University of Indonesia* **2012–2014**
◇ Delivered faculty-assigned lectures, engineered assessment materials, and supervised teaching assistants. Contributed to research initiatives within the Formal Methods in Software Engineering laboratory.

EDUCATION

PHD IN COMPUTER SCIENCE.
Czech Technical University (ČVUT). **2025–present**

PHD IN MATHEMATICS.
University of Canterbury (NZ). **2014–2019**

MASTER OF COMPUTER SCIENCE.
University of Indonesia. **2011–2012**

BACHELOR OF SCIENCE IN MATHEMATICS
University of Indonesia. **2006–2009**

PUBLICATIONS

1. Anggha Nugraha. *Paraconsistent Dominated Convergence*. In arXiv:2607.15723 [math.LO], 2026.
2. Anggha Nugraha. *A Constructive Field of Infinitesimals: Chunk and Permeate Approach*.¹ In arXiv:2607.08206 [math.LO], 2026.
3. Stefan Ratschan, Anggha Nugraha, Mikoláš Janota, Marek Dančo. *Inductive Satisfiability Certification for Universal Quantifiers and Uninterpreted Function Symbols*. In arXiv:2602.16335 [cs.LO], 2026.
4. Anggha Nugraha, M. McKubre-Jordens, and H. Diener. *Naïve Infinitesimal Analysis: Its Construction and Its Properties*. In arXiv.2009.11424 [math.LO], 2020.

5. T. Antaresti, Anggha Nugraha, et al. *Wavelet denoising and fractal feature selection for classifying simulated earthquake signals from mobile phone accelerometers*. In IEEE 11th International Multi-Conference on Systems, Signals & Devices (SSD 14), 2014.
6. M. Arzaki and Anggha S. Nugraha. *Translation of an Allen's Temporal Logic Variant to RPNL*. In Journal of Mathematical and Computational Science (3), 2013, pages 1405-1429.
7. A. S. Nugraha and T. Basaruddin. *Analysis and Comparison of QR Decomposition Algorithms for Some Types of Matrix*. In Proceedings of Federated Conference on Computer Science and Information System (FedCSIS), 2012, pages 561-565.

¹Updated and comprehensive version; supersedes the 2020 preprint (arXiv:2009.11424) and the PhD thesis.

CONFERENCES AND TALKS

- ◊ Alpine Verification Meeting 2026: *Inductive Satisfiability Certification for Universal Quantifiers and Uninterpreted Function Symbols*. Eger, Hungary
- ◊ New Zealand Mathematics and Statistics Postgraduate Conference in 2014, 2017, and 2018. Auckland, New Zealand
- ◊ Various informal talks in The Netherlands (especially at the University of Amsterdam) when doing research collaboration with Prof. Jan Bergstra in 2016.
- ◊ 7th Non-Classical Logic: Theory and Applications conference in 2015. Toruń, Poland
- ◊ 1st National Joint Seminar in Mathematics at the University of Indonesia in 2010: *The Generating Function Applications for Counting The Number of Pattern Graph from Main Pattern Graph*.

GRANTS AND SCHOLARSHIPS

- ◊ LPDP Scholarship from the Ministry of Finance of The Republic of Indonesia 2014-2019
Scholarship to do my PhD study at the University of Canterbury, NZ.
- ◊ The New Zealand Mathematical Society (NZMS) Travel Grant 2016
Travel grant to do research collaboration with Prof. Jan Bergstra of the University of Amsterdam, The Netherlands.
- ◊ The New Zealand Mathematical Society (NZMS) Travel Grant 2015
Travel grant to present my paper at "The 7th Conference on Non-Classical Logic: Theory and Application" in Poland.
- ◊ IMHERE (Indonesia Managing Higher Education for Relevance and Efficiency) Scholarship for Master Study 2011-2012

REFERENCES

1. doc. Dipl.Ing. Stefan Ratschan, Dr.-tech.
Institute of Computer Science, Czech Academy of Sciences
stefan.ratschan@cs.cas.cz
2. Dr. Maarten McKubre-Jordens
School of Mathematics and Statistics University of Canterbury
mkm@papanui.school.nz
3. Prof. Dr. Kiki Ariyanti Sugeng
Department of Mathematics University of Indonesia
kiki@sci.ui.ac.id
4. Phillipa Gourdie (Senior Tutor)
School of Mathematics and Statistics University of Canterbury